

IE 509 Lab 10 Data Visualization Assignment

Use the wine dataset to answer the following questions.

<https://gist.githubusercontent.com/tijptjik/9408623/raw/b237fa5848349a14a14e5d4107dc7897c21951f5/wine.csv>

Data Visualization Tasks

1. Create a histogram to visualize the distribution of `Alcohol` content in the wines.
2. Generate a box plot to show the distribution of `Proline` content across wines.
3. What is the correlation between various chemical attributes in the dataset?
4. Create a pair plot for `Malic.acid`, `Ash`, and `Proanth`.
5. What is the relationship between `Ac1` content and `Color.int` levels in the wines? Create a scatter plot to visualize this relationship.
6. How does the average `Flavanoids` content differ by wine type? Calculate and plot the average `Flavanoids` grouped by the `Wine` feature.
7. Create a violin plot of `Proanth` by the `Wine` type.

Clustering and Dimensionality Reduction

8. Visualize clusters using K-Means clustering in 2D scatter plots for all pairwise combinations of `Alcohol`, `Proline`, and `Phenols` from the Wine dataset. Determine the suitable number of clusters for this data. Once the number of clusters has been identified, visualize the clusters by plotting them, using different colors or markers to represent each cluster.
9. Project the high-dimensional wine data to two dimensions using t-SNE. Create a scatter plot of the resulting two-dimensional data points. Use different colors or markers to represent different wine types from the `Wine` feature.
10. Perform PCA on the wine data to reduce the dimensions to two principal components. Create a scatter plot of the two principal components. Use different colors or markers to represent different wine types from the `Wine` feature.
11. Create a plot showing the explained variance ratio for each principal component.
12. Create a cumulative variance plot to show the total variance explained as a function of the number of principal components.

13. Calculate the 25th, 50th (median), and 75th percentiles for the features (**Alcohol**, **Malic.acid**, **Phenols**, **Proline**) in the dataset. Create a box plot to visually represent the distribution, including the 25th, 50th, and 75th percentiles. Clearly label the quartiles on the box plot.